

Fourier analysis – 4

Discrete Fourier analysis and characters

Homework

13 сентября 2026 года

Theoretical minimum. The characters of a finite abelian group form an orthonormal basis of the space of functions on it, so every function has a unique expansion and Parseval holds. Convolution becomes multiplication, which turns counting problems — how many solutions of an equation, how many subsets with a prescribed sum — into estimates of $\widehat{f}$. The root-of-unity filter is the case $G = \mathbb{Z}_n$ and extracts terms with exponent divisible by n from a generating function. Character sums of quadratic type have modulus $\sqrt{p}$, which is the exact strength one needs in most applications. Finally, smallness of all non-trivial coefficients means that the function behaves like its average, and this is the mechanism behind every statement of the form “an almost homomorphism is close to a homomorphism” or “a set without structure contains the expected number of patterns”.

1. Prove that $\widehat{\widehat{f}}(x) = Nf(-x)$ on $\mathbb{Z}_N$, and compute the transform of the indicator of an arithmetic progression and of the indicator of the set of quadratic residues modulo a prime.
2. Prove that the number of representations $n = a + b$ with $a \in A$, $b \in B$ for $A, B \subseteq \mathbb{Z}_N$ equals $\frac{1}{N} \sum_r \widehat{\mathbf{1}_A}(r) \widehat{\mathbf{1}_B}(r) e^{2\pi i r n / N}$, and deduce that $|A| + |B| > N$ forces $A + B = \mathbb{Z}_N$.
3. Prove the Cauchy–Davenport theorem $|A+B| \geq \min(p, |A|+|B|-1)$ in $\mathbb{Z}_p$ by any method you like, and explain why a purely Fourier-analytic proof is hard (compare with the polynomial method of Polynomials – 8).

4. Let χ be a non-trivial multiplicative character modulo p . Prove that $|\sum_x \chi(x) e^{2\pi i a x/p}| = \sqrt{p}$ for $a \neq 0$, and deduce Gauss' evaluation of the quadratic sum up to sign (the modulus was computed in Trig-2).
5. Prove that the number of solutions of $x^2 + y^2 \equiv a \pmod{p}$ is $p - \left(\frac{-1}{p}\right)$ for $a \neq 0$, by writing the count as a convolution of the transform of the squares.
6. Prove the discrete uncertainty principle $|\text{supp } f| \cdot |\text{supp } \widehat{f}| \geq N$ for every non-zero $f: \mathbb{Z}_N \rightarrow \mathbb{C}$, show that the indicator of a subgroup gives equality, and prove the sharper bound $|\text{supp } f| + |\text{supp } \widehat{f}| \geq p+1$ when $N = p$ is prime.
7. Prove that for odd N the chirp $f(x) = e^{2\pi i x^2/N}$ satisfies $|f| \equiv 1$ and $|\widehat{f}(r)| = \sqrt{N}$ for every r , and deduce once more the value $|G| = \sqrt{p}$ of the quadratic Gauss sum.
8. Let $A \subseteq \mathbb{Z}_N$. Prove the identity $\sum_{r \neq 0} |\widehat{\mathbf{1}_A}(r)|^2 = |A|(N - |A|)$, deduce that $\max_{r \neq 0} |\widehat{\mathbf{1}_A}(r)| \geq \sqrt{\frac{|A|(N - |A|)}{N - 1}}$, and describe the sets for which this bound is attained.
9. Prove that the $N \times N$ matrix $F = (e^{2\pi i j k/N})_{j,k}$ satisfies $F\overline{F} = NE$, compute $\det F$ up to sign, and find the eigenvalues of F together with their multiplicities.
10. Deduce the Poisson summation formula on $\mathbb{Z}_N$: for a subgroup $H \leq \mathbb{Z}_N$ and any f , $\sum_{x \in H} f(x) = \frac{|H|}{N} \sum_{r \in H^\perp} \widehat{f}(r)$, and use it to count the elements of a subgroup lying in a given arithmetic progression.